

SHEPHERD BAY, NU, Canada

WMO: 719110

Lat: 68.817N

Lon: 93.433W

Elev: 140

StdP: 14.62

Time Zone: -7.00 (NAM)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-52.4	-49.4	-50.6	0.3	-40.8	-46.3	0.4	-30.9	36.8	-24.4	31.5	-20.2	7.0	140	1.133

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	64.5	55.3	60.6	53.0	56.9	50.8	55.1	61.7	52.6	58.4	50.2	55.1	11.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.8	57.3	57.2	49.0	51.6	55.4	46.5	47.0	53.2	23.3	62.0	21.8	58.9	20.4	55.4	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.7	27.7	24.6	-55.7	70.5	4.3	4.4	-58.8	73.6	-61.3	76.2	-63.8	78.7	-66.9	81.9
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-30.0	-31.3	-22.2	-5.9	14.1	34.4	47.4	43.8	32.4	14.6	-7.7	-21.5
	DBStd	29.79	12.09	11.41	12.78	11.95	9.51	7.55	6.86	6.50	6.35	10.49	12.92	13.43
	HDD50	16182	2480	2277	2237	1678	1112	474	135	214	529	1097	1732	2217
	HDD65	21576	2945	2697	2702	2128	1577	919	546	657	979	1562	2182	2682
	CDD50	81	0	0	0	0	0	5	55	21	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.7	10.8	10.5	10.9	11.3	11.9	11.3	11.7	12.0	13.0	14.2	11.9	10.8
Precipitation	PrecAvg	6.1	0.2	0.2	0.2	0.3	0.3	0.6	0.9	1.4	1.0	0.7	0.3	0.2
	PrecMax	9.5	0.7	1.0	1.2	1.2	0.7	2.4	2.6	3.5	2.5	1.7	0.7	0.5
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.2	0.0	0.0
	PrecStd	1.7	0.2	0.2	0.3	0.3	0.2	0.6	0.6	0.8	0.6	0.4	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	4.8	15.1	26.4	33.8	63.6	70.2	66.5	52.9	35.3	25.1	17.3
		MCWB	9.7	4.2	14.3	25.6	32.8	52.7	58.5	57.5	49.7	33.9	24.7	16.9
	2%	DB	0.2	-1.2	8.9	21.8	32.4	55.7	66.2	61.7	48.0	32.5	20.6	11.3
		MCWB	-0.5	-1.9	8.0	20.7	31.7	49.1	54.7	55.7	44.9	32.0	19.7	10.7
	5%	DB	-5.5	-6.7	4.1	17.2	30.3	50.1	63.0	57.7	44.1	30.4	15.8	5.2
		MCWB	-6.1	-7.3	3.5	16.4	29.4	46.1	54.0	52.1	42.3	29.9	15.1	4.5
	10%	DB	-11.5	-13.3	-1.2	12.8	27.5	45.0	59.7	54.3	41.1	28.2	11.5	-0.8
		MCWB	-12.1	-13.8	-1.8	12.2	26.6	42.0	51.9	50.2	39.5	27.5	10.8	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	5.4	10.3	24.3	32.5	52.9	58.9	60.7	49.4	35.4	22.1	13.4
		MCDB	10.6	6.1	11.1	25.1	32.9	62.3	70.2	66.3	52.3	36.4	22.6	13.6
	2%	WB	0.9	-1.5	5.1	18.2	30.4	50.1	55.2	56.2	44.5	31.5	15.1	10.1
		MCDB	1.4	-0.9	5.7	18.8	31.0	57.2	63.1	61.4	46.7	31.8	15.8	10.6
	5%	WB	-4.3	-8.0	-0.1	14.1	27.0	46.4	52.7	53.2	41.3	28.8	10.4	5.7
		MCDB	-3.6	-7.6	0.3	14.8	27.7	50.9	59.8	57.3	42.5	29.3	11.0	6.4
	10%	WB	-11.1	-14.6	-6.1	10.0	24.1	42.2	50.4	50.3	38.0	25.3	5.2	-3.5
		MCDB	-10.5	-14.2	-5.6	10.7	24.8	44.7	56.1	54.0	39.3	25.9	5.7	-3.0
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.7	14.3	15.7	11.9	8.3	13.0	10.8	7.1	10.1	12.8	13.4
		MCDBR	18.3	18.3	16.6	15.1	10.0	15.3	18.7	16.8	12.5	9.3	14.8	15.9
		MCWBR	17.6	17.1	15.1	16.5	12.1	9.9	10.7	11.5	10.2	8.4	13.9	14.1
	5% WB	MCDBR	18.3	17.4	17.0	15.0	12.3	15.0	17.5	15.6	11.5	8.7	14.5	14.1
		MCWBR	18.1	17.0	16.7	14.9	12.2	9.9	10.8	10.8	10.2	9.1	14.5	14.0
Clear-Sky Solar Irradiance	taub		0.149	0.207	0.251	0.304	0.321	0.326	0.351	0.320	0.293	0.233	0.159	N/A
	taud		1.886	2.054	2.178	2.074	2.121	2.344	2.390	2.509	2.472	2.108	1.890	N/A
	Ebn at Noon		29	187	250	263	272	276	264	262	238	166	19	0
	Edh at Noon		2	14	24	36	39	33	30	24	19	13	1	0
All-Sky Solar Radiation	RadAvg		8	145	635	1402	2046	1902	1640	1077	532	167	26	0
	RadStd		1	9	25	42	85	175	153	96	46	18	2	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.16	+2.16	N/A	N/A	N/A	N/A	-785	-778	N/A	N/A
(47) Regional (5 neighbors)		+1.28	+1.60	N/A	N/A	N/A	N/A	-542	-526	N/A	N/A

Nomenclature: See separate page